

Samkit Bothra

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EDUCATION

Florida Atlantic University, Boca Raton, FL

August 2023 - Present

Dual-enrolled Bachelor of Science in Computer Science

Minor in Business

- 3.8 /4.0 GPA

Florida Atlantic University High School, Boca Raton, FL

May 2026

Florida Atlantic University High School is an extremely competitive and rigorous program where students are fully dual-enrolled at Florida Atlantic University from tenth grade through twelfth grade; students take college courses to fulfill high school requirements. Most students graduate from FAU High School with 80-90 college credits.

- 3.86/4.0 GPA

LEADERSHIP

Student Mayor, Mayor, Parkland Florida

September 2024 – Present

- Serving as the official student representative of the city, advocating for youth perspectives and collaborating with city leaders on initiatives that impact local communities.
- Planning, hosting, and participating in citywide events, fostering community engagement, and promoting civic involvement among residents.
- Developing public speaking, project management, and leadership skills through active involvement in city government activities and public forums.

Jain Center of South Florida (IRCC) Youth Group, President

December 2024 - Present

- Plan and oversee the execution of large-scale celebrations for major Indian holidays, attracting over 500 attendees.
- Design and deliver interactive lessons on Hindu traditions and values, fostering cultural awareness, pride, and community engagement among children.
- Developing strong leadership, project management, and communication skills

INSTRUCTIONAL EXPERIENCE AND MENTORING

Physics Learning Assistant, Math Learning Center, FAU, FL

August 2025 – Present

- Supported instruction in an undergraduate physics course of ~100 students, circulating the classroom to assist with problem-solving and concept clarification.
- Guided students through challenging physics topics, fostering critical thinking and collaborative learning.
- Partnered with faculty to implement active learning strategies that improved student engagement and performance.

OURI Peer Mentor, Office of Undergraduate Research & Inquiry, FAU, FL

August 2025 - Present

- Mentoring undergraduate students in attaining research opportunities, funding applications, and submitting to conferences and journals.
- Specializing in the College of Engineering and Computer Science.
- Fostering a supportive research community by sharing best practices and encouraging scholarly engagement across disciplines.

- College Mathematics Tutor**, Math Learning Center, FAU, FL January 2025 – May 2025
- Providing one-on-one and small-group tutoring for college-level algebra, enhancing students' foundational understanding and problem-solving abilities.
 - Tailoring tutoring sessions to accommodate diverse learning styles, integrating real-life applications and interactive exercises that foster engagement and mastery of complex concepts.
 - Guided students in developing study plans, time management skills, and test-taking strategies.

- FAU High Peer Mentor**, FAU, Boca Raton, FL August 2024 – May 2025
- Mentored high school underclassmen in navigating Florida Atlantic University's dual enrollment process, offering personalized guidance on registration, course selection, and university protocols.
 - Provided ongoing consultation and support to address academic difficulties, including resolving issues with professors and managing class requirements.
 - Developed and refined key skills in communication, problem-solving, and mentoring, ensuring students receive the necessary tools and confidence to succeed.

- STAR Lab Instructor**, FAU High School, *Boca Raton, Florida* January 2023 -Present
- Instructing young researchers and assisting them in their projects
 - Taught younger researchers about basic lab skills and etiquette

- Kumon**, Coral Springs, FL August 2023 – January 2024
- Helped young kids to improve their math and reading skills (be more specific)
 - Gained instructional experience in teaching and instructing

CONFERENCES AND SYMPOSIUMS

- BristleCon 2025**, San Francisco, *California* January 22 - January 24, 2025
- Engaged with industry-leading experts from Sofar Ocean, Bristlemouth, and other prominent organizations in ocean exploration and technology.
 - Presented research project on Underwater Acoustic Sensing and Wireless Communications to CEOs and startup founders, aiming to secure strategic partnerships for further innovation.
 - Collaborated with senior engineers at Sofar Ocean and Bristlemouth, receiving valuable feedback and brainstorming enhancements to advance the project's technological potential.

- Florida Undergraduate Research Conference**, USF, Tampa, *Florida* February 14 – February 15, 2025
- Presented “Exploring Gender Differences in Perceptions of AI’s Impact on Social Media” at the Florida Undergraduate Research Conference (FURC), a study examining how men and women differ in their views on misinformation, privacy, and the societal influence of AI-driven content.
 - Highlighted the growing importance of understanding demographic perceptions in the age of AI to inform ethical tech development and responsible social media design.
 - Connected with experts in the field and received vital insights on how to further develop and expand the project.

- FAU OURI Symposium**, FAU, Boca Raton, *Florida* April 4, 2025
- Gave a 12-minute oral presentation to judges on my AI’s influence on misinformation in social media polling project.
 - Awarded a first-place certificate along with a \$100 cash prize for the importance and presentation style of the project.
 - Also held a poster presentation on my biomechanics project titled “Validation of a Wearable IMU Sensor against the Optotrak Motion Capture Device for Assessment of Head and Neck Tracking”

- National Conference on Undergraduate Research**, Pittsburgh, *Pennsylvania* April 7 – April 9, 2025

- Delivered a formal oral presentation on my research analyzing gender-based differences in public perceptions of AI's influence on social media, misinformation, and privacy
- Focused on the methodology and statistical findings of a 16-question survey distributed to over 200 respondents
- Connected with experts from all around the nation in various fields and received vital insights on how to further develop and expand the project.

PROFESSIONAL DEVELOPMENT

Young Scholars Program, Member, Florida State University, Tallahassee, Florida June 7 – July 19, 2025

- One of the 40 students selected into this six-week STEM summer program out of a pool of hundreds of students across Florida.
- Took classes such as Quantum Computing and Game Theory
- Worked with an FSU professor as a Molecular Biophysics Researcher

National Honor Society, Member, FAU High School, Boca Raton, Florida October 2023-Present

- Went to several beach cleanups to raise awareness of pollution.
- Learned about college scholarships and what is required to receive them.
- Helped others to apply for and research which scholarships they would want to apply to.

Key Club, Secretary, FAU High School, Boca Raton, Florida September 2023 - Present

- Organizing volunteer events such as food for the poor.
- Presenting to members of the club and getting them to participate in volunteer activities.

ACADEMIC PROJECTS

MD Simulations to Uncover Cryptic Sites in SARS-CoV-2 Protease, YSP, FSU June 7 – July 19, 2025

- Performed molecular dynamics simulations of the SARS-CoV-2 main protease to identify cryptic allosteric pockets not visible in static structures.
- Using enhanced sampling techniques, we ran a 300 ns simulation and successfully identified a transient pocket at the dimer interface.
- This hidden pocket could serve as a novel drug target for future COVID-19 antiviral development, offering a new strategy to inhibit viral replication.
- Tools used:
 - Linux
 - Visual Molecular Dynamics
 - CHARMM-GUI

Underwater Acoustic Sensing and Wireless Communications, CAAI, FAU June 2023 - Present

- Conducting a cutting-edge research project on Underwater Acoustic Sensing and Wireless Communications, aimed at enhancing ocean surveying methods and data accuracy.
- Delving into technical innovation by integrating advanced sensor arrays, signal processing algorithms, and robust software solutions to enable reliable, real-time underwater data transmission.
- Presented key findings at a high-profile conference, engaging directly with CEOs and leading experts in ocean exploration and technology.
- Tools used:
 - WinSCP
 - Python
 - C Programming
 - Linux
 - Raspberry PI

- Terminal Command Prompt

Wearable Sensors for Tracking Body Motion: Advancing Rehabilitation, FAU

January 2025 – Present

- Developed and tested a wearable IMU-based system to track head and neck movement for assessing fall risk and postural control in patients
- Used AI-driven analysis and 3D modeling to reconstruct movement patterns and visualize deviations in the center of mass over time
- Validated sensor data by comparing it against the gold-standard Optotrak motion capture system using vector angle comparisons.
- Tools used:
 - Python
 - Lenovo 3D-camera software
 - Excel
 - Optotrak
 - First Principles

Business Economic Polling Initiative, Florida Atlantic University

August 2024 – Present

- Conducted an in-depth survey-based research study on “Exploring Gender Differences in Perceptions of AI’s Impact on Social Media,” aiming to uncover how diverse genders perceive AI’s growing societal relevance.
- Collected and analyzed data from over 200 respondents, refining statistical analysis and data visualization skills to draw meaningful insights on user attitudes and experiences.
- Tools Used:
 - Statistical Package for the Social Sciences (SPSS)

Summer Lab Skills Workshop, Pompano Beach, FL 33062

June 19-23, 2023

- Conducted field surveys by gathering microplastics from Wahoo Bay.
- Gained skills in teamwork through working with a team of college graduates to collect data.
- Gained skills in leadership through leading a team through sample analysis.

PUBLICATIONS

- Samkit Bothra, Edward P. Davis, Valentina N. Saracino, Alpa Raval, Rebecca Millard, Vibhor Agrawal, Abhijit S. Pandya, "A Brief Overview: The Utility of Wearable Motion Sensor Technology for Neurodegenerative Diseases," International Journal of Applied Sciences & Development, vol. 4, pp. 205-213, 2025, DOI:10.37394/232029.2025.4.22

RELEVANT COURSEWORK

Research Methods, Business Economic Polling Initiative (BEPI), Data Structures & Algorithms, Computer Logic Design, Physics for Engineers 1 & 2, Calculus 1, 2 & 3, Financial Accounting, Managerial Accounting, Statistics

SKILLS

Programming Languages

- Python, C++, C

Technical Skills

- Linux
- Raspberry PI software integration
- Cellular data transmission
- Data Science Fundamentals

- Underwater Robotics

Languages

- **English** (100% Proficiency), **Hindi** (100% Proficiency), **Spanish** (50% Proficiency)

HONORS AND AWARDS

Awarded 1st Place award and \$100 Cash Prize at FAU OURI Symposium

- Developed a survey deployed to individuals across the country.
- Gave a 12-minute oral presentation to judges on this AI's influence on misinformation in social media polling project.

Distributive Education Clubs of America International Qualifier

- Placed third in the entire state in March 2025 for exam + roleplay performance and qualified for the International Career Development Conference (DECA ICDC).
- Qualified for states in January 2025 with a first-place award in district event for the season, with a 90 on the exam and a 98 on the roleplay, and recognition from the school for the highest average in the chapter.
- Excelled in role-play acting as a financial advisor and gaining skills in communication and business

Several Debate Tournament Wins

- Participated and placed in several district wide debate tournaments in Lincoln Douglas.
- Gained many communication skills through speaking in front of big crowds.

Two-time State Golf Qualifier and Individual District Champion

- Playing on the high school varsity golf team at the number 3/4 spot on the five-team roster.
- Played several individual tournaments and placed 1st and 2nd three times in South Florida .